Build Butterbear extras

1. Data + routes / Nemo

Favorites, special requests, booking email jobs,
and a reminder that includes Open-Meteo weather.

00:00–00:30 • Synthetic narration

Store favorites on the user

src/models.js + src/app.js

favorites: [{ type: ObjectId, ref: 'Campground' }]

$addToSet: { favorites: campground._id }
$pull: { favorites: campgroundId }

Save a special request

src/models.js + booking handlers

```
specialRequest: { type: String, default: '',
                  maxlength: 2000 }

text(req.body.specialRequest ?? '',
     'specialRequest', 2000, { required: false })
```

Validate the booking dates

src/validation.js / bookingDates()

```
nights = end.diff(start, 'days').days
1 <= nights && nights <= 3

arrival = start.set({ hour: 14 })
reminder = start.minus({ days: 1 }).set({ hour: 9 })
```

Persist the booking and email jobs

src/app.js / POST /bookings

```
Booking.create({ ...input,
  notifications: notifyInput(input, campground,
    dates, 'confirmation', 1, req.user.email)
})

201  { emailStatus: 'queued', ... }
```

Handle edits and admin access

src/app.js / PATCH /bookings/:id

```
filter = { _id, ...bookingOwner(req), status: 'active' }

findOneAndUpdate({ ...filter, revision: current.revision }, {
  $set: { ...input, revision: current.revision + 1 },
  $push: { notifications: { $each: newJobs } }
})
```

Claim and deliver notification jobs

src/services/notifications.js / deliverNext()

pending → processing → sent

claimToken + leaseUntil + attempts
check booking status + reminder revision
sendMail(composeEmail(booking, job, forecast))

Fetch weather for the check-in date

src/services/weather.js / getWeather()

GET https://api.open-meteo.com/v1/forecast
latitude, longitude, timezone, start_date, end_date

daily: weather_code, temperature_2m_max,
temperature_2m_min, precipitation_probability_max

Send real email through Gmail

src/services/mail.js / createMailer() + composeEmail()

MAIL_TRANSPORT=smtp
SMTP_HOST=smtp.gmail.com
SMTP_PORT=587
SMTP_SECURE=false

SMTP_USER and SMTP_PASSWORD stay in ignored .env

Demonstrate and explain the result

Postman Runner / Butterbear Final Presentation

1 Core steps 01-12 → class + sequence diagrams
2 Extra credits step 15 → Gmail inbox

Show confirmation, update, and weather reminder.
Code: github.com/chanichkorn/butterbear